TEST DESIGN SPECIFICATION (TDS)

1. Requirements

RQ-001: The system shall allow librarians to add a new book by providing a valid title, author, and ISBN.

RQ-002: The system shall allow librarians to borrow books and record borrowing transactions.

2. Feature Sets

FS-001 – Book Management

Objective: Manage book information.

Priority: High

Traceability: RQ-001

FS-002 – Loan Management

Objective: Manage borrowing transactions.

Priority: High

Traceability: RQ-002

3. Test Conditions

ID	Feature Set	Description	Priority	Traceability
TCond-001	FS-001	Add book with valid title, author, and ISBN	High	RQ-001
TCond-002	FS-001	Add book with invalid ISBN format	High	RQ-001

TCond-003	FS-001	Add book with duplicate ISBN	Medium	RQ-001
TCond-004	FS-002	Borrow available book	High	RQ-002
TCond-005	FS-002	Borrow unavailable book	High	RQ-002
TCond-006	FS-002	Borrow book with incomplete borrower information	Medium	RQ-002

4. Coverage Items

ID	Description	Related Test Condition	Priority	Traceability
CI-001	Title field	TC-001	High	RQ-001
CI-002	Author field	TC-001	High	RQ-001
CI-003	ISBN format validation	TC-002	High	RQ-001
CI-004	Duplicate ISBN validation	TC-003	Medium	RQ-001
CI-005	Book availability	TC-004	High	RQ-002
CI-006	Borrower information	TC-006	Medium	RQ-002

5. Test Traceability Matrix (TTM)

The following example demonstrates the application of a Test Traceability Matrix (TTM) based on the Test Design Specification presented in this document.

Requirement	Feature Set	Test Condition	Coverage Item

RQ-001	FS-001	TCond-001	CI-001, CI-002
RQ-001	FS-001	TCond-002	CI-003
RQ-001	FS-001	TCond-003	CI-004
RQ-002	FS-002	TCond-004	CI-005
RQ-002	FS-002	TCond-005	CI-005
RQ-002	FS-002	TCond-006	CI-006

6. Test Design Technique Summary

Test Condition ID	Test Design Technique	Rationale
TCond-001	Equivalence Partitioning	Classify valid and invalid categories for title, author, and ISBN inputs.
TCond-002	Boundary Value Analysis	Verify ISBN length limits and boundary values.
TCond-003	Equivalence Partitioning	Distinguish duplicate and non-duplicate ISBN categories.
TCond-004	Equivalence Partitioning	Separate valid and invalid book availability conditions.
TCond-005	Equivalence Partitioning	Verify borrowing requests for unavailable books.
TCond-006	Equivalence Partitioning	Classify complete and incomplete borrower information.